

Understanding Kubernetes version, node image the upgrade and why we need to upgrade

About Kubernetes versions

- A **Kubernetes version** is a release of the Kubernetes software with **new features, fixes and improvements** from **upstream Kubernetes** or the **cloud provider (Azure)**
- The **format** of a Kubernetes version is **[major].[minor].[patch]**
 - 1.25.5:
 - 1 – major; 25 – minor; 5 – patch
- **Major** change when backwards compatibility or incompatible **API updates** might be **broken**
- **Minor** change when backwards-compatible functionality **changes** are **made to the other minor versions**
- **Patch** versions change when **bugs** that are backwards compatible are **fixed**

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Kubernetes versions upgrade in AKS

- At any given time, **AKS supports N-2 minor versions**
- **You cannot skip minor versions, except:**
 - Only when **upgrading from an unsupported version back into the minimum supported version** is it possible to skip multiple versions
- Control planes and nodes have Kubernetes versions
 - **Major** version of the **control plane and node pool must be the same**
 - **Node pool** version **cannot be higher** than the control plane version
 - The **control plane** can have a **minor** Kubernetes version higher **than** the **node pool**, but with **only two increments**
- You can upgrade:
 - Control plane and node pool(s) **at the same time**
 - **Separately**, control plane and then the node pool(s)

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Upgrade operation in AKS

- **Buffer node(s) are added** during the operation
- Default max surge values:
 - Node pools added **before** Kubernetes version 1.27: **1**
 - Node pools added **after** Kubernetes version 1.27: **10%**
- Existing **nodes** are **cordoned** and **drained**
- The **buffer nodes run the workload of the existing nodes**
- **Existing nodes are reimages/recreated** with the newer Kubernetes version
- **Max-surge dictates the speed** of the upgrade operation

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Node images in AKS

- **Updated node images** include updated **kernel updates**, **Kubernetes security updates**, and **component version changes**
- Node images are updated **weekly** for **Linux** and **monthly** for **Windows**
- When you **upgrade the node image**, you will get the **latest** one
- Node image is **about the OS**, like Ubuntu or Mariner
 - AKSUbuntu-1804gen2containerd-2023.02.15
 - AKSCBLMariner-V2gen2-2023.02.15
- After **Kubernetes version upgrade**, **latest node image will be brought** for the nodes

Why we need to upgrade

- New features
- Improvements
- Bug fixes
- Security patches
- Receive support